ASSIGNMENT-AR/VR

The task was to make a basic AR application by placing a globe onto a marker. When camera is opened, by pointing the custom marker I created to the camera, the globe appears to sit directly on top of it.

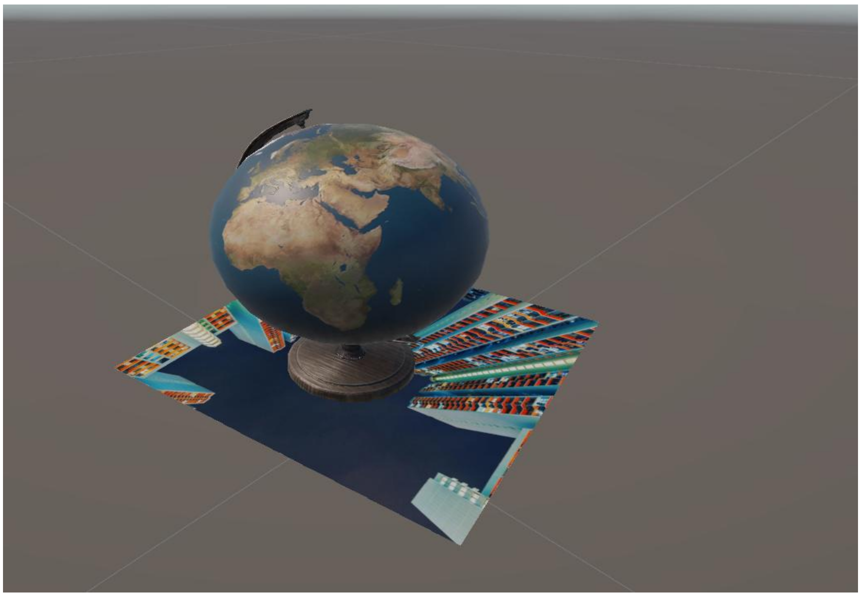
This task uses a 3d globe. For handling 3d assets like this, I used unity. For AR tracking, I used Vuforia engine.

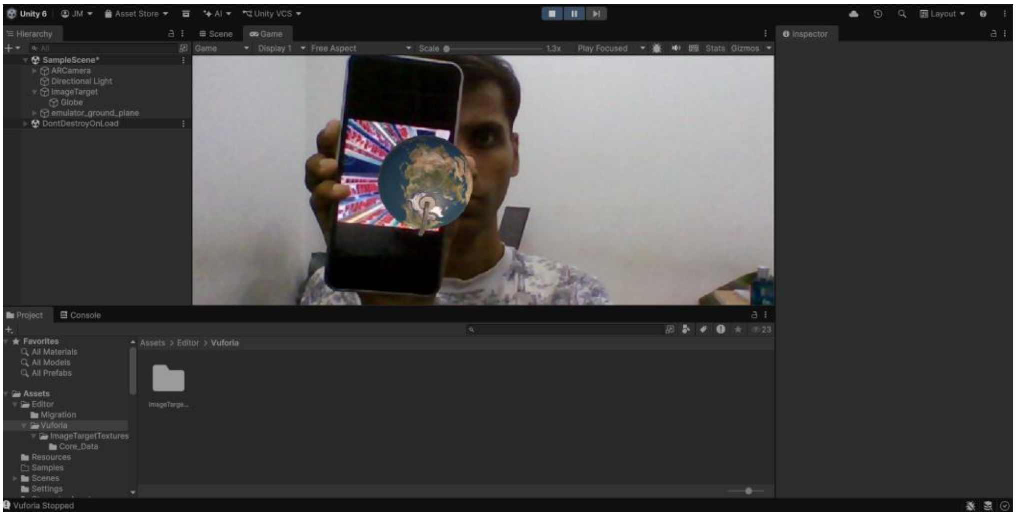
Vuforia Engine step by step

1. First, I created a Vuforia Developer account. In the License Manager, I generated a free "Development Key," which acts as the handshake between Unity and the Vuforia servers.
2. I chose a custom image with high "feature contrast" (lots of edges and unique patterns). This is crucial because Vuforia uses these details to calculate the camera's position.
3. In the Vuforia Target Manager, I created a new database and uploaded my marker image. After the system processed it, I checked the "Star Rating" which I got 5 stars.
4. Finally, I downloaded the database for import to unity

Unity step by step

1. From unity asset store, I installed 3d globe model and imported to my unity project
2. I imported the Vuforia Engine package and my custom database into my unity project.
3. I replaced the standard Main Camera with the **ARCamera** prefab. In the Inspector, I pasted my License Key into the Vuforia Configuration settings.
4. I added an **Image Target** prefab to the scene. In its settings, I selected my database and the specific marker I uploaded earlier.
5. The globe I imported placed onto the **Image Target**, making it a child object of it.
6. When I placed it, the globe shaders were not loading. So I used Universal Render Pipeline (URP), I converted the globe’s shaders from "Built-in" to "URP/Lit" and manually re-linked the globe’s texture (Albedo map) to fix the pink appearance.(Used gemini for this fix)
7. I hit "Play" and held my custom marker up to my webcam to verify the globe appeared.




Unity 6
JM
Asset Store
AI
Unity VCS
Layout
Hierarchy
Scene
Game
Inspector
Game
Display 1
Free Aspect
Scale
1.3x
Play Focused
Stats
Gizmos
SampleScene*
ARCamera
Directional Light
ImageTarget
Globe
emulator_ground_plane
DontDestroyOnLoad
Project
Console
Favorites
All Materials
All Models
All Prefabs
Assets
Editor
Migration
Vuforia
ImageTargetTextures
Core_Data
Resources
Samples
Scenes
Settings
Assets > Editor > Vuforia
Vuforia Stopped